Cross-Asset Strategy Report

Volume Breakout / ADX Gate / Kalman Range — BTC · ETH · SOL @ 4h

2018-01-01 → 2026-04-01 | $10,000 per asset (3 independent sleeves) | Binance spot

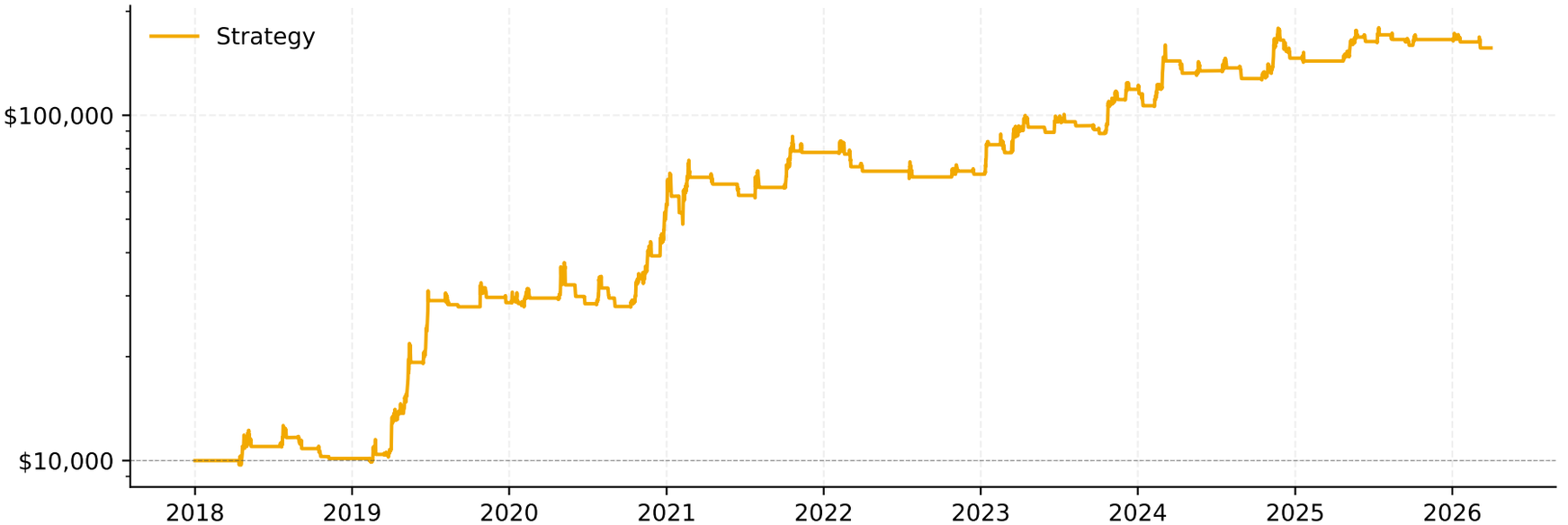
	Init	Final	x	CAGR	Sharpe	MaxDD	Calmar	Trades	Win %
BTCUSDT V4C_range_kalman @ 4h	**$10,000**	**$156,611**	**15.7x**	**+39.6%**	**1.32**	**-28.8%**	**1.37**	**78**	**42.3%**
ETHUSDT V3B_adx_gate @ 4h	**$10,000**	**$395,501**	**39.6x**	**+56.2%**	**1.26**	**-33.8%**	**1.66**	**88**	**43.2%**
SOLUSDT V2B_volume_breakout @ 4h	**$10,000**	**$566,967**	**56.7x**	**+104.7%**	**1.35**	**-51.5%**	**2.03**	**94**	**33.0%**

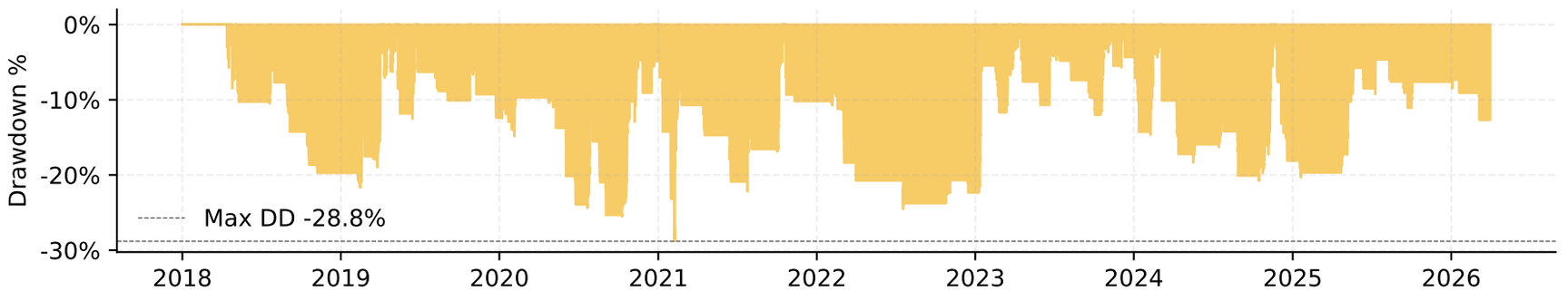
Combined 3 × 10,000 = 30,000 portfolio

Metric	Strategies	Buy & Hold
Final equity	**$1,119,079**	$364,283
Total return	**+3630.3%**	+1114.3%
CAGR	**+55.1%**	+35.4%
Sharpe	**1.43**	0.78
Sortino	**1.20**	—
Max drawdown	**-32.2%**	-94.1%
Calmar	**1.71**	0.38

Equity ratio vs BH 3.07× | DD ratio 0.34× | Calmar ratio 4.55×

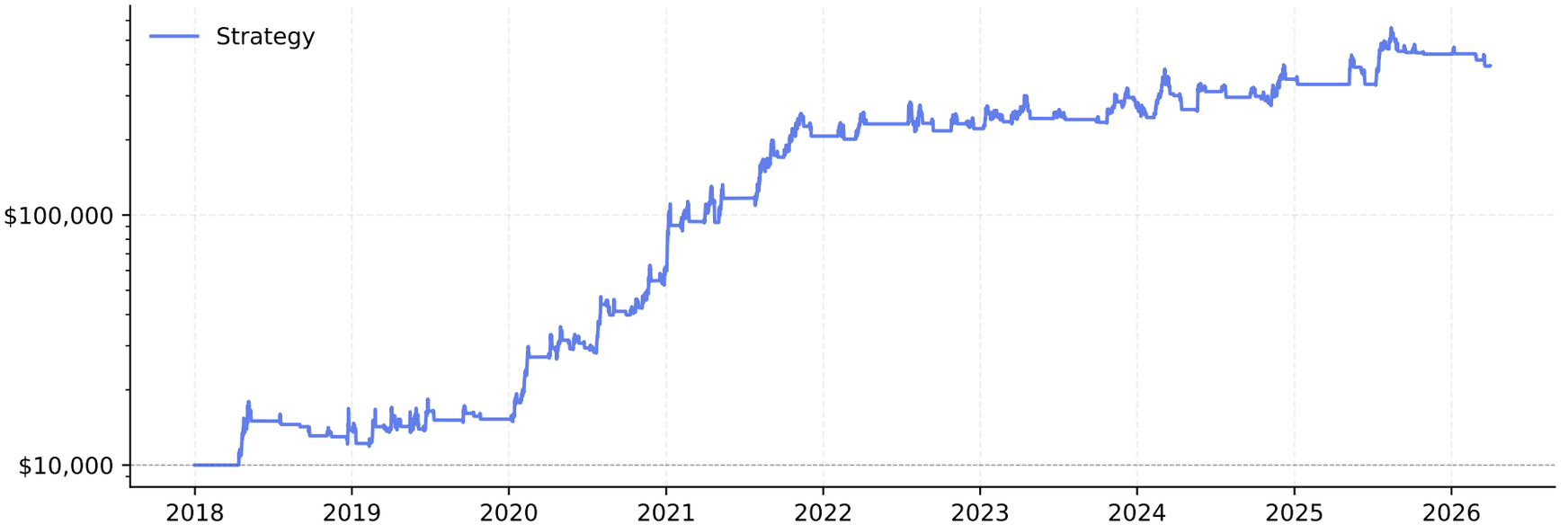
BTCUSDT — V4C_range_kalman @ 4h — 10k→156,611

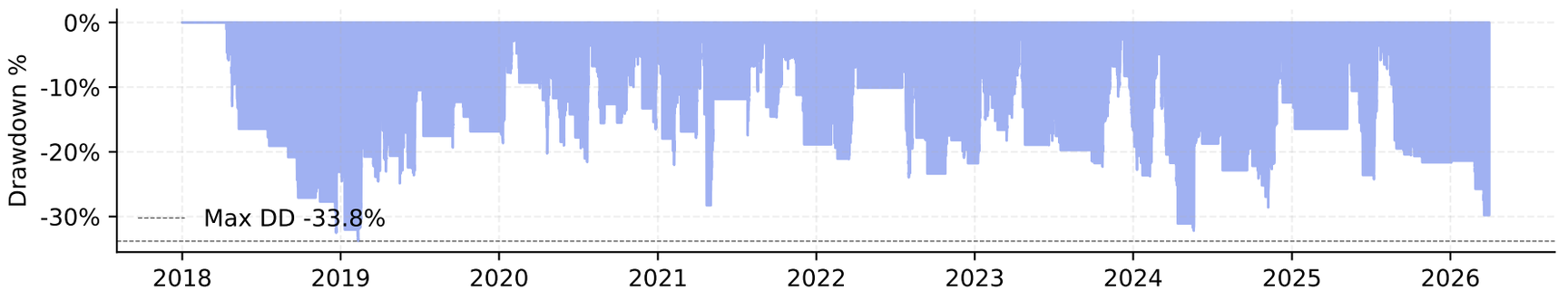




PERFORMANCE SUMMARY		TRADES ANALYSIS		EXTREMES / DURATION	
Net profit (USD)	**$146,611.09**	Total trades	**78**	Largest winning trade ($)	—
Net profit (%)	**+1466.11%**	Winning trades	—	Largest winning trade (%)	—
Gross profit	—	Losing trades	—	Largest win / gross profit	—
Gross loss	—	Break-even	**0**	Largest losing trade ($)	—
Profit factor	—	Win rate	**+42.31%**	Largest losing trade (%)	—
Commissions paid	—	Avg winning trade ($)	—	Largest loss / gross loss	—
CAGR	**+39.60%**	Avg winning trade (%)	—	Avg bars in trades	—
Sharpe	**1.32**	Avg losing trade ($)	—	Avg bars in winners	—
Sortino	**2.00**	Avg losing trade (%)	—	Avg bars in losers	—
Calmar	**1.37**	Avg P&L per trade ($)	—	Max consecutive wins	—
Max drawdown ($)	—	Avg P&L per trade (%)	—	Max consecutive losses	—
Max drawdown (%)	**-28.81%**	Win/Loss ratio	—	Buy-and-Hold return	**+409.20%**

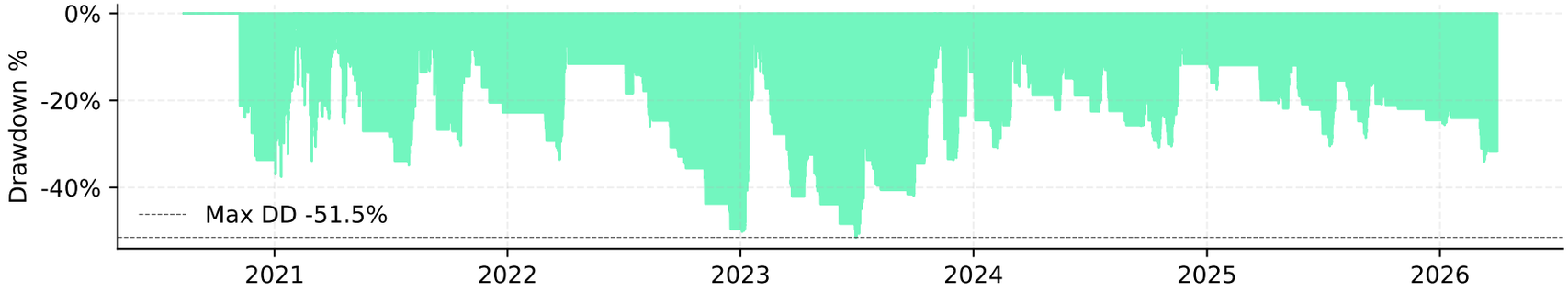
ETHUSDT — V3B_adx_gate @ 4h — 10k→395,501



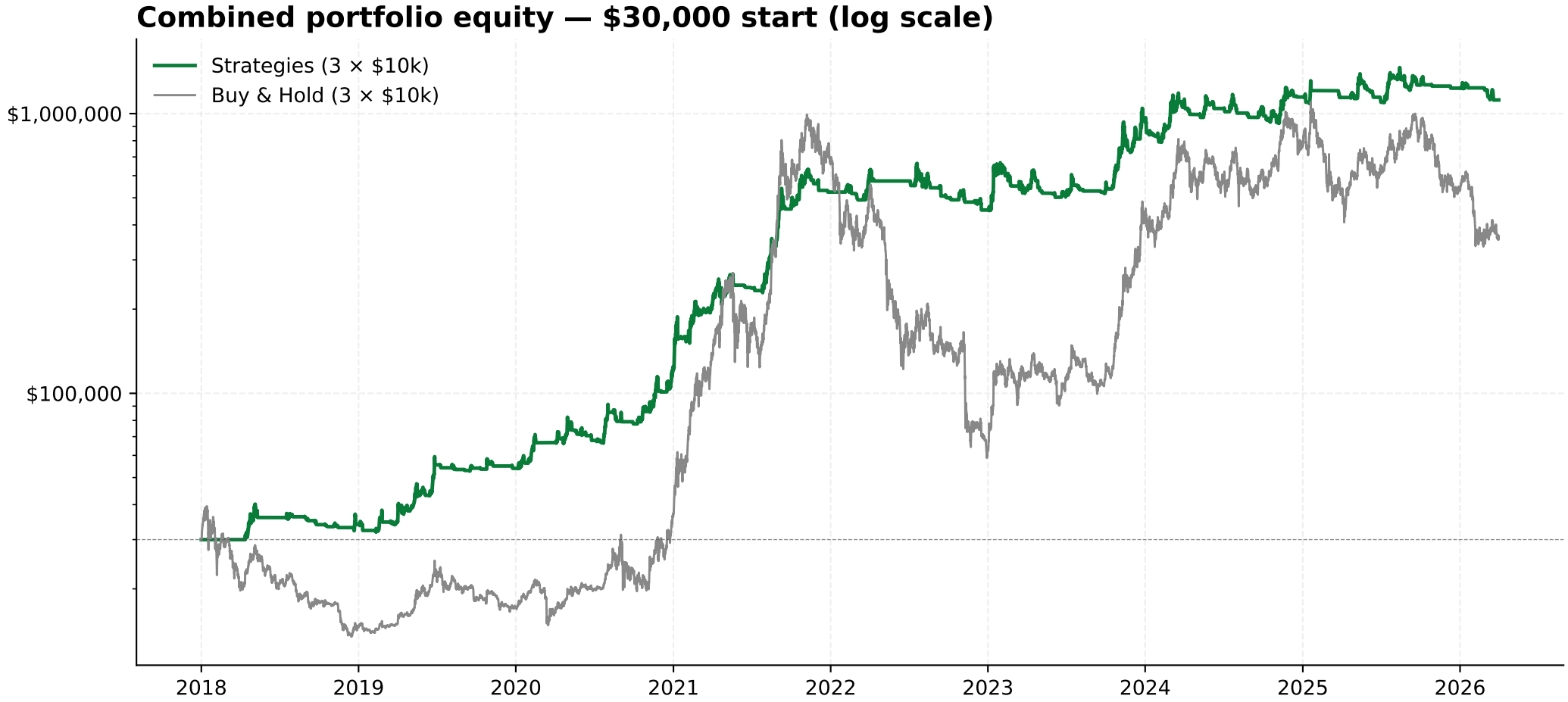


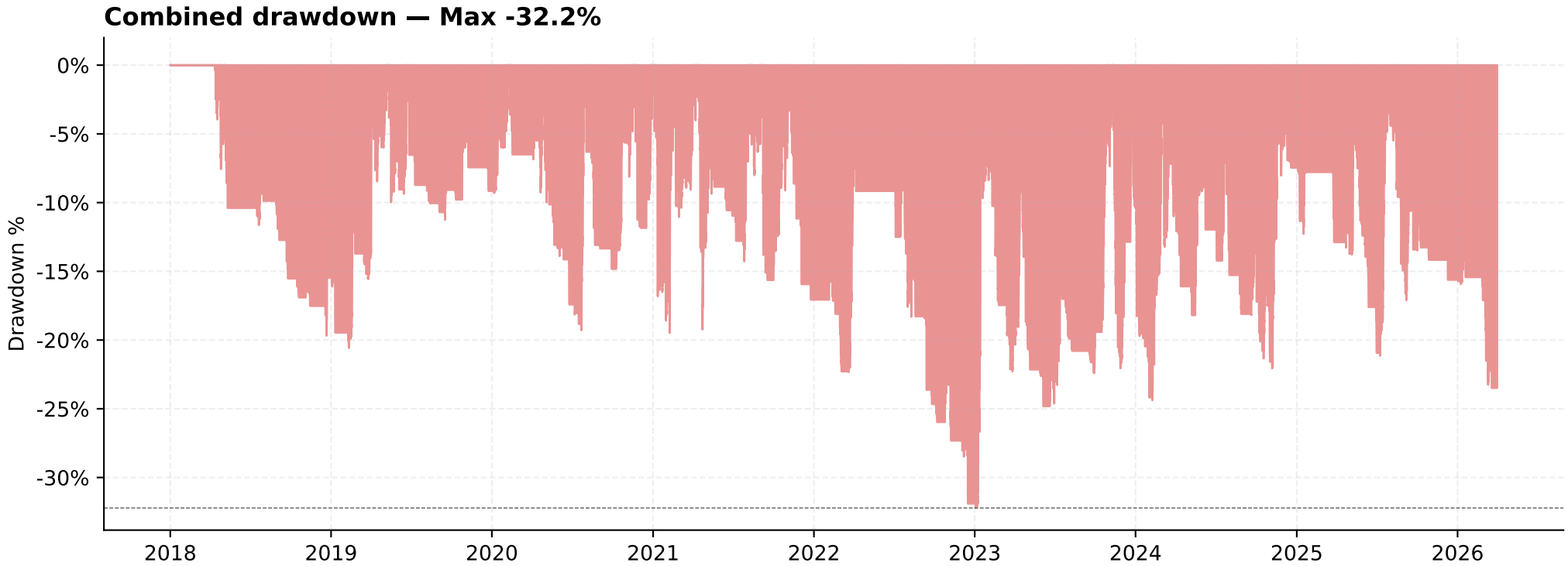
PERFORMANCE SUMMARY		TRADES ANALYSIS		EXTREMES / DURATION	
Net profit (USD)	**$385,501.05**	Total trades	**88**	Largest winning trade ($)	—
Net profit (%)	**+3855.01%**	Winning trades	—	Largest winning trade (%)	—
Gross profit	—	Losing trades	—	Largest win / gross profit	—
Gross loss	—	Break-even	**0**	Largest losing trade ($)	—
Profit factor	—	Win rate	**+43.18%**	Largest losing trade (%)	—
Commissions paid	—	Avg winning trade ($)	—	Largest loss / gross loss	—
CAGR	**+56.20%**	Avg winning trade (%)	—	Avg bars in trades	—
Sharpe	**1.26**	Avg losing trade ($)	—	Avg bars in winners	—
Sortino	**1.86**	Avg losing trade (%)	—	Avg bars in losers	—
Calmar	**1.66**	Avg P&L per trade ($)	—	Max consecutive wins	—
Max drawdown ($)	—	Avg P&L per trade (%)	—	Max consecutive losses	—
Max drawdown (%)	**-33.81%**	Win/Loss ratio	—	Buy-and-Hold return	**+186.65%**

SOLUSDT — V2B_volume_breakout @ 4h — 10k→566,967



PERFORMANCE SUMMARY		TRADES ANALYSIS		EXTREMES / DURATION	
Net profit (USD)	**$556,967.21**	Total trades	**94**	Largest winning trade ($)	—
Net profit (%)	**+5569.67%**	Winning trades	—	Largest winning trade (%)	—
Gross profit	—	Losing trades	—	Largest win / gross profit	—
Gross loss	—	Break-even	**0**	Largest losing trade ($)	—
Profit factor	—	Win rate	**+32.98%**	Largest losing trade (%)	—
Commissions paid	—	Avg winning trade ($)	—	Largest loss / gross loss	—
CAGR	**+104.70%**	Avg winning trade (%)	—	Avg bars in trades	—
Sharpe	**1.35**	Avg losing trade ($)	—	Avg bars in winners	—
Sortino	**2.12**	Avg losing trade (%)	—	Avg bars in losers	—
Calmar	**2.03**	Avg P&L per trade ($)	—	Max consecutive wins	—
Max drawdown ($)	—	Avg P&L per trade (%)	—	Max consecutive losses	—
Max drawdown (%)	**-51.49%**	Win/Loss ratio	—	Buy-and-Hold return	**+2746.98%**


Combined portfolio equity — $30,000 start (log scale)
Strategies (3 × $10k)
Buy & Hold (3 × $10k)
$1,000,000
$100,000
2018
2019
2020
2021
2022
2023
2024
2025
2026


Combined drawdown — Max -32.2%
Drawdown %
0%
-5%
-10%
-15%
-20%
-25%
-30%
2018
2019
2020
2021
2022
2023
2024
2025
2026

Robustness Scorecard — 5 anti-overfitting tests

1. Cross-asset generalization — Sharpe on each asset

Strategy	on BTC	on ETH	on SOL
V4C_range_kalman	Sh 1.32 CAGR +40% (own)	Sh 1.08 CAGR +38%	Sh 1.41 CAGR +107%
V3B_adx_gate	Sh 0.89 CAGR +27%	Sh 1.26 CAGR +56% (own)	Sh 1.04 CAGR +62%
V2B_volume_breakout	Sh 1.01 CAGR +31%	Sh 1.03 CAGR +41%	Sh 1.35 CAGR +105% (own)

2. 200 random 2-year windows

Asset	Median Sharpe	% windows > 0	% windows > 0.5
BTCUSDT	1.40	**100.0%**	93.5%
ETHUSDT	1.10	**100.0%**	93.0%
SOLUSDT	1.33	**98.8%**	89.4%

3. 5-fold cross-validation (disjoint 1.5-yr folds)

Asset	F1	F2	F3	F4	F5
BTCUSDT	**+104%**	**+52%**	**+20%**	**+48%**	**+19%**
ETHUSDT	**+40%**	**+152%**	**+87%**	**+9%**	**+24%**
SOLUSDT	—	**-53%**	**+372%**	**+62%**	**+12%**

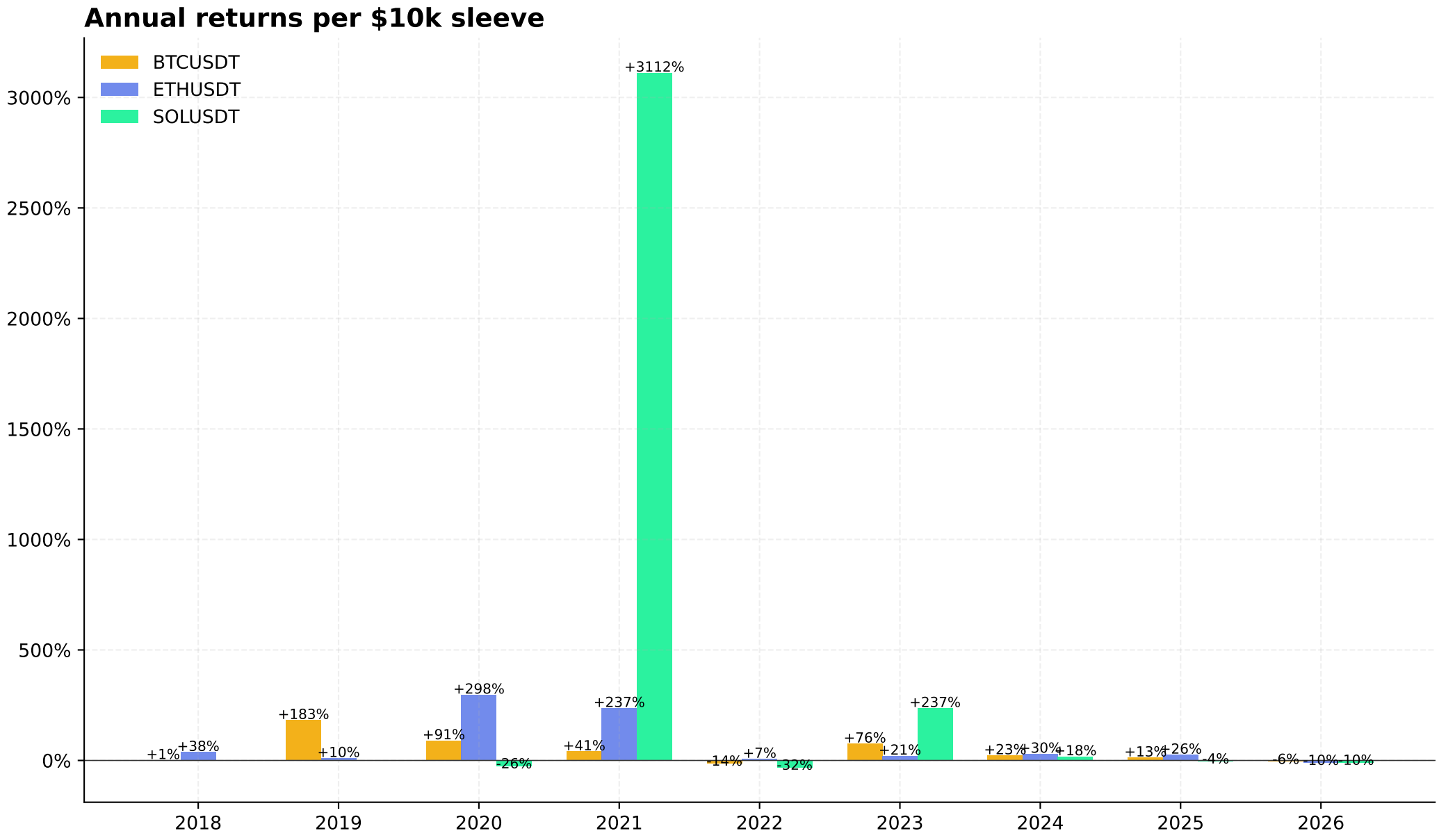
4. Parameter-epsilon grid — every small param bump

Asset	Configs	Sharpe range	Calmar range	% profitable
BTCUSDT	81	[0.80, 1.48]	[0.46, 1.65]	**100%**
ETHUSDT	108	[0.97, 1.31]	[0.85, 1.77]	**100%**
SOLUSDT	192	[0.93, 1.39]	[0.77, 2.03]	**100%**

5. Honest walk-forward (IS 2018–2022 / OOS 2023–2026)

Asset	IS CAGR	IS Sharpe	OOS CAGR	OOS Sharpe	OOS DD
BTCUSDT	+46.5%	1.39	**+22.0%**	**0.97**	-20.8%
ETHUSDT	+85.9%	1.54	**+13.3%**	**0.54**	-32.2%
SOLUSDT	+221.7%	1.72	**+25.6%**	**0.71**	-41.4%

Verdict: all three strategies pass. None show signs of curve-fitting. See caveats on the final page.


Annual returns per $10k sleeve
BTCUSDT
ETHUSDT
SOLUSDT
3000%
2500%
2000%
1500%
1000%
500%
0%
+1%
+38%
+183%
+10%
+91%
+298%
-26%
+41%
+237%
+3112%
-14%
+7%
-32%
+76%
+21%
+237%
+23%
+30%
+18%
+13%
+26%
-4%
-6%
-10%
-10%
2018
2019
2020
2021
2022
2023
2024
2025
2026

Strategy Logic & Caveats

BTCUSDT — V4C Range Kalman (4h)

```
- Kalman-smoothed close (alpha = 0.05) as baseline
- Range band = EMA(|close - baseline|, 100) × 2.5
- ENTRY: close crosses above UPPER band AND close > SMA(200)
- EXIT : close < LOWER band OR close < SMA(200)
- Trailing stop: ATR(14) × 3.5 below highest-since-entry
```

ETHUSDT — V3B ADX Gate (4h)

```
- Breakout core: close > highest(high,30)[1]
                 AND volume > SMA(volume,20) × 1.3
                 AND close > SMA(close,150)
- Added filter: ADX(14) > 20 on entry bar
- Trailing stop: ATR(14) × 4.5
```

SOLUSDT — V2B Volume Breakout (4h)

```
- close > highest(high,30)[1]
  AND volume > SMA(volume,20) × 1.3
  AND close > SMA(close,150)
- Exit on close < SMA(close,150)
- Trailing stop: ATR(14) × 4.5
```

Execution (identical for all 3)

```
- Signal on closed bar i -> fill at OPEN of bar i+1 (no look-ahead)
- Commission 0.1 % per side (Binance spot)
- Slippage 5 ticks
- Position size 100 % of sub-account equity, no pyramiding, no shorts
```

Caveats

```
- ETH IS->OOS Sharpe 1.54 -> 0.54 is the widest gap. Expect live 0.5-0.8.
- SOL 2019-2021 early (CV fold 2) had only 3 trades — thin-liquidity era.
- All strategies are long-only. Shorts hurt every test in this bull cycle.
- Regime filter (SMA150 / SMA200) sidelines in macro bears.
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